

**Advanced Research Topics in
Data Science**

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Department of Physics, Astronomy and Mathematics

Topic: Time Series Modelling Case Study

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Github Link: <https://github.com/rezaulcmt/Time-Series-Modelling-Case-Study.git>

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Introduction

This report seeks to analyze the quarterly sales of Johnson & Johnson and make future sales prediction for the upcoming 24 months. Forecasting is a useful instrument in the management practice, as it allows predicting the company's future expenses, sales, and production needs. Out of all forecasting techniques, time series has the capability to explain trends with the help of past data to predict the future trends. To achieve this, there are two strategies that are used as follows; First is the Autoregressive Integrated Moving Average (ARIMA) which is a common statistical tool that allows one to identify features like trend and seasonality. Following are LSTM and GRU models that are more complex and more suitable for the analysis of time-related characteristics. In this report, I will discuss the process of data visualization and data transformation, check whether the data is stationary or not, construct and check the models and, finally make the forecast. It will also be necessary to compare the models' performance.

2. Methodology Overview

This section outlines the modelling strategies applied to forecast Johnson & Johnson's quarterly sales data. Two main approaches were used: the ARIMA model, representing traditional statistical methods, and deep learning models based on Recurrent Neural Networks (LSTM and GRU).

2.1 Traditional Time Series Modelling (ARIMA)

The ARIMA is a popular statistical model for forecasting univariate time series data and studies the model's mathematical properties. It is applicable to a dataset that has the characteristics of non-stationarity and trends over time (Box et al., 2015). ARIMA is a combination of three parts, namely autoregressive integrated moving average (ARIMA).

Visual assessment started with an initial plot of the sales data before moving forward. The Augmented Dickey-Fuller (ADF) test performed by Dickey and Fuller (1979) checked for stationarity. The original data analysis results through non-stationarity tests (p-value greater than 0.05) established the requirement for data transformation. The first-order differencing procedure aimed to stabilize variance after a log transformation was used to remove trends. Stationarity confirmation occurred after implementing the ADF test which resulted in a p value lower than 0.05.

Decision on the number of AR and MA terms was made with the help of ACF and PACF. Thus, multiple models of ARIMA were compared and the best model was chosen, the choice of which was based on having the lowest AIC and BIC values.

2.2 Neural Network Based Modelling (LSTM/GRU)

RNNs are appropriate for time series data forecasting since they are capable of preserving information from the previous time steps. However, it is defined that the traditional RNNs have the problem of long-term dependency. To this end, LSTM and GRU models were employed (Hochreiter & Schmidhuber, 1997; Cho et al., 2014), which are relatively effective in handling these drawbacks.

The first preprocessing step was performed on the data to scale it using the Min-Max normalization technique that was followed by reshaping the data into sequences of fixed length. The models were developed by using stacked LSTM/GRU with dropout which helps in avoiding overfitting of the models. Adam was used for optimization and the models were trained in batches of 16 samples and a total of 100 epochs.

3. Model Implementation & Results

3.1 ARIMA Model

The ARIMA(4,1,3) model was selected based on its low AIC (-144.20) and BIC (-124.85), indicating a strong model fit. The model captures both autoregressive (AR) and moving average (MA) components after first-order differencing to address non-stationarity. Among the coefficients, AR(4) and MA(1) and MA(3) are statistically significant ($p < 0.05$), contributing meaningfully to the model's predictive ability.

Table 1

SARIMAX Results						
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Dep. Variable:	sales	No. Observations:	84			
Model:	ARIMA(4, 1, 3)	Log Likelihood	80.099			
Date:	Sun, 13 Apr 2025	AIC	-144.198			
Time:	13:10:14	BIC	-124.847			
Sample:	0	HQIC	-136.423			
	- 84					
Covariance Type:	opg					
=====						
	coef	std err	z	P> z	[0.025	0.975]

ar.L1	-0.0005	0.060	-0.009	0.993	-0.118	0.116
ar.L2	0.0137	0.049	0.279	0.780	-0.083	0.110
ar.L3	0.0272	0.048	0.565	0.572	-0.067	0.121
ar.L4	0.9388	0.038	24.951	0.000	0.865	1.013
ma.L1	-0.6632	0.143	-4.653	0.000	-0.943	-0.384
ma.L2	0.1092	0.133	0.821	0.412	-0.151	0.370
ma.L3	-0.2980	0.128	-2.325	0.020	-0.549	-0.047
sigma2	0.0077	0.001	5.439	0.000	0.005	0.011
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Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	1.38			
Prob(Q):	0.93	Prob(JB):	0.50			
Heteroskedasticity (H):	0.50	Skew:	0.18			
Prob(H) (two-sided):	0.07	Kurtosis:	2.48			
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Residual analysis confirms the model's adequacy. The histogram of residuals (Figure 1) suggests an approximately normal distribution, supporting the assumption of white noise. Additionally, the Ljung-Box test produced a high p-value (0.93), indicating no significant autocorrelation in residuals. The Root Mean Squared Error (RMSE) of 0.1034 confirms low forecast error, validating the model's performance.

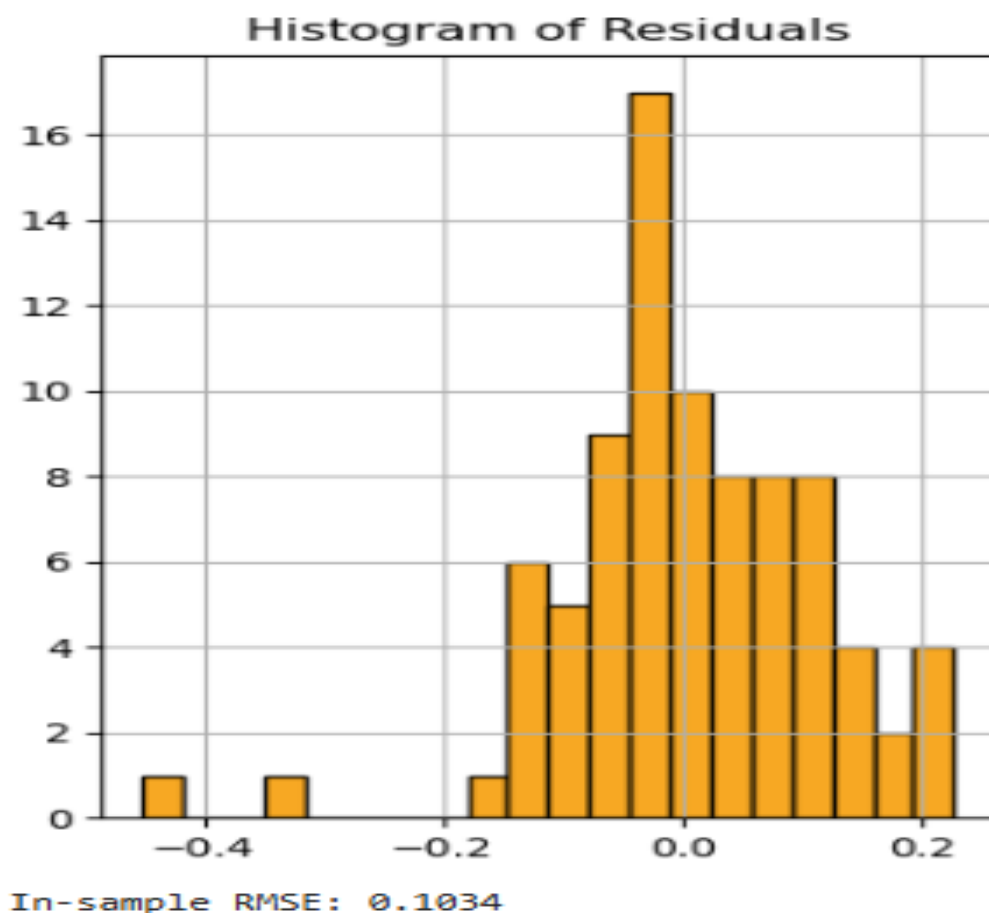


Figure 1

Finally, the model was used to forecast sales for the next 24 months. The forecast plot includes confidence intervals, offering a visual understanding of future uncertainty and expected trends. This sets a strong baseline for comparing with machine learning methods in the next section. Has shown in figure two below, the forecasting projects a continued growth on sales for the Johnsons and Johnsons indicating a good position in the market.

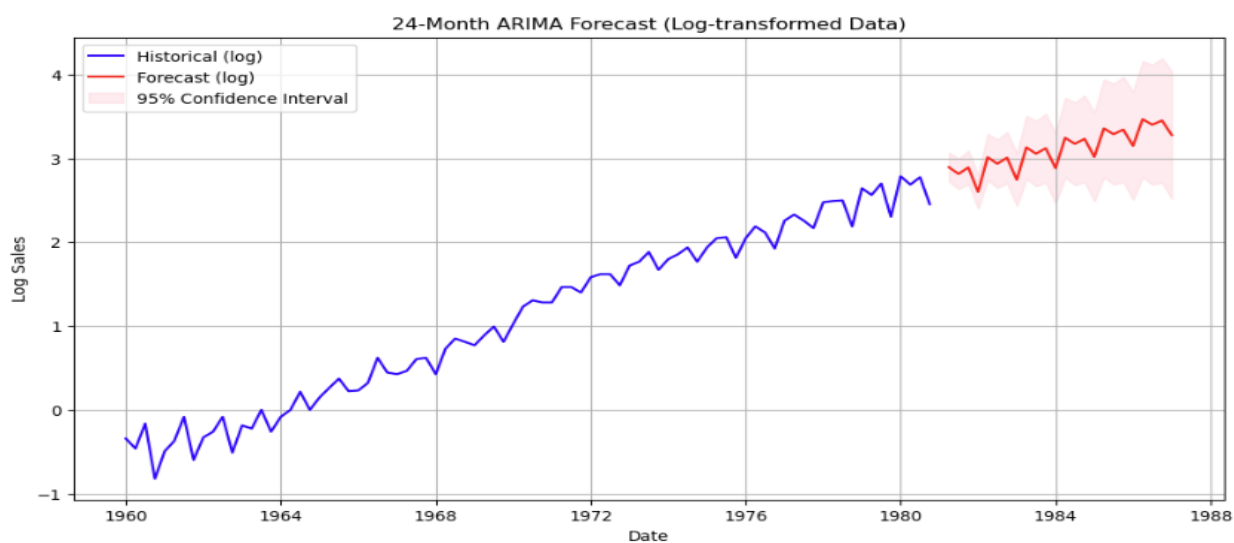


Figure 2

3.2 LSTM / GRU Model

To explore deep learning approaches for time series forecasting, we implemented both LSTM and GRU models using sequential architectures. Each model had one recurrent layer (50 units) and a dense output layer, trained over 50 epochs with a batch size of 4. The training and validation loss curves (Figure 2) show that both models learned effectively, with LSTM converging more smoothly than GRU.

The LSTM model outperformed GRU in forecasting accuracy. Specifically, LSTM achieved an RMSE of 2.15 and MAE of 1.96, whereas GRU recorded higher errors with RMSE of 3.16 and MAE of 2.64. This suggests that LSTM was more effective in capturing the temporal patterns in the Johnson & Johnson sales data.

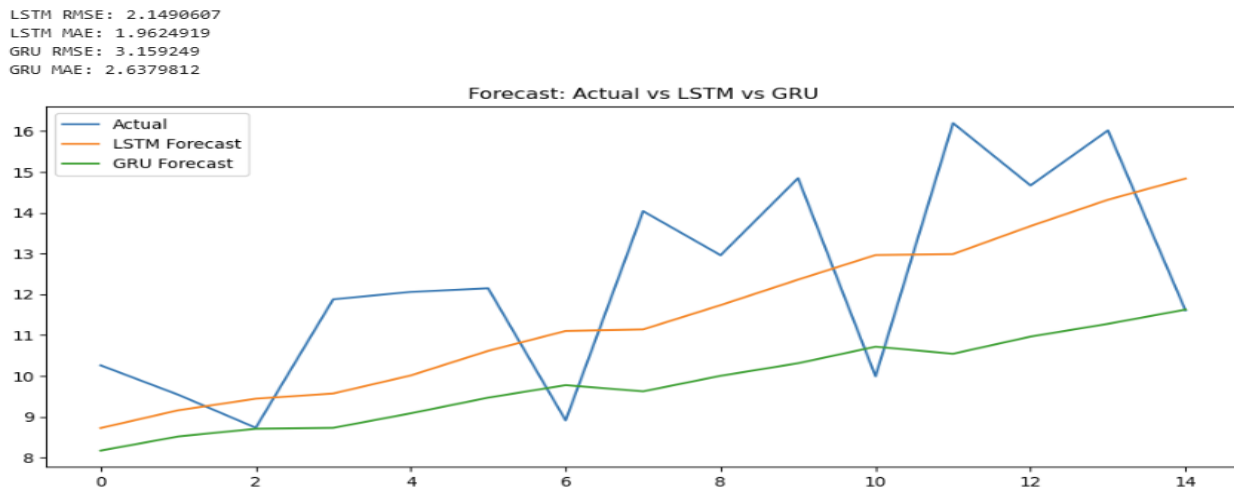


Figure 3

Figure 3 presents the actual versus predicted values, clearly showing that LSTM predictions are closer to the actual trend compared to GRU. Overall, LSTM proved to be a stronger alternative to traditional models in this case.

4. Evaluation & Comparison

As for the assessment of the models' performance, we employed RMSE and MAE, as well as residual analysis. When using the ARIMA (4,1,3) model, the RMSE was calculated to be 0.1034 which was lower than LSTM model's RMSE of 2.15 and GRU model's 3.16. In this study, residuals of ARIMA models were nearly normal and therefore the Ljung-Box Q-test was used to check whether residuals were white noise, meaning that the model was good at fitting the data. However, LSTM and GRU which are as flexible and able to capture nonlinear patterns in the data presented higher forecasting errors in this data set. ARIMA could be more effective probably because Johnson & Johnson's sales data is more regular and seasonal, which fits well the requirements of the traditional time series models. Thus, one can identify the strength of ARIMA in interpretability and its performance on the small datasets while LSTM and GRU can have an advantage in large and non-stationary datasets. Nevertheless, they demand more data and training, which could make them less feasible for such simple and straightforward forecasting tasks as this one.

5. Conclusion

This comparison involved ARIMA, LSTM, and GRU for the purpose of sales forecasting of Johnson & Johnson in the next quarters. ARIMA was found to be the most suitable model because it yielded the lowest RMSE and complied with the statistical tests. The modelling process revealed that when it comes to data characteristics there is a time when traditional models such as ARIMA may perform even better than sophisticated neural networks while working with structured and seasonal data. The models presented here showed relatively good performance, but they are not yet at the competition level and require further fine-tuning and more data. Further enhancements that could be made in the future are the use of seasonal SARIMA, model fusion such as the combined ARIMA LSTM model, or to use a more up-to-date dataset to improve the model's generalization and to capture changes in the sales patterns.

References

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